ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ-3 **SUMMER SEMESTER, 2024-2025**

DURATION: 40 MINUTES **FULL MARKS: 15**

Math 4641: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

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1. In Figure 1, consider the shaded regions A and B, enclosed between the two functions $f(x)$ and $g(x)$, where the region B is bounded on the right by the line $x = 3.75$. Suppose the positive difference of the areas of the shaded regions is represented by d, which means $d = A - B$, and it is known that $d = 3.5$ square units. The aforementioned functions are, 5 (CO2) (PO1)

$$f(x) = x^2 - kx$$
$$g(x) = x + \sin(\pi x)$$

Using the Composite Trapezoidal Rule, with $n = 10$ segments, approximate the value of k.
(Note: Your calculator must be in radian mode.)

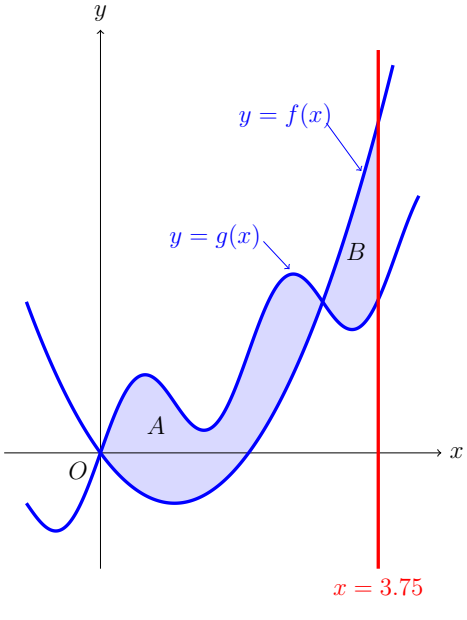


Figure 1: The shaded regions A and B.

Solution:

Let the two curves cross at $x = c$, where $0 < c < 3.75$, i.e., the abscissa at which the regions A and B meet. Since $g(x)$ lies above $f(x)$ before the crossing and below it afterwards,

$$A = \int_0^c [g(x) - f(x)]\,dx, \qquad B = \int_c^{3.75} [f(x) - g(x)]\,dx$$

Hence, taking the positive difference of the two areas flips the sign of the second integral, and the two pieces merge into a single one,

$$d = A - B = \int_0^c [g(x) - f(x)]\,dx + \int_c^{3.75} [g(x) - f(x)]\,dx = \int_0^{3.75} [g(x) - f(x)]\,dx \quad (1)$$

so that the crossing abscissa c is never required. The integrand is,

$$F(x) = g(x) - f(x) = (1 + k)x - x^2 + \sin(\pi x)$$

We can approximate the integral value of a function $f(x)$ using composite trapezoidal rule with n segments, each of width $h = \dfrac{b-a}{n}$, between the lower-limit a and the upper-limit b as follows,

$$\int_a^b f(x)dx = \frac{h}{2}\left[f(a) + 2\left\{\sum_{i=1}^{n-1} f(a + ih)\right\} + f(b)\right]$$

Hence, we can expand Equation 1 by applying this formula, with $n = 10$ and $h = \dfrac{3.75 - 0}{10} = 0.375$,

$$d \approx \frac{h}{2}\left[F(0) + 2\left\{\sum_{i=1}^{9} F(0 + ih)\right\} + F(3.75)\right]$$

The sine terms, evaluated in radians, are,

$$\begin{aligned}
\sin(0.375\pi) = \sin(2.625\pi) &= 0.9239, & \sin(0.75\pi) = \sin(2.25\pi) &= 0.7071,\\
\sin(1.125\pi) = \sin(1.875\pi) &= -0.3827, & \sin(1.5\pi) &= -1,\\
\sin(3\pi) &= 0, & \sin(3.375\pi) &= -0.9239,\\
\sin(3.75\pi) &= -0.7071
\end{aligned}$$

Therefore, the nodal values of the integrand are,

$$\begin{aligned}
F(0) &= 0, & F(0.375) &= 0.375k + 1.1583, & F(0.75) &= 0.75k + 0.8946,\\
F(1.125) &= 1.125k - 0.5233, & F(1.5) &= 1.5k - 1.75, & F(1.875) &= 1.875k - 2.0233,\\
F(2.25) &= 2.25k - 2.1054, & F(2.625) &= 2.625k - 3.3417, & F(3) &= 3k - 6,\\
F(3.375) &= 3.375k - 8.9395, & F(3.75) &= 3.75k - 11.0196
\end{aligned}$$

Substituting these into the expansion,

$$\begin{aligned}
d &\approx 0.1875\big[0 + 2 \times \big(16.875k - 22.6304\big) + \big(3.75k - 11.0196\big)\big]\\
&= 0.1875\big[37.5k - 56.2804\big]\\
&= 7.0313k - 10.5526
\end{aligned}$$

Finally, equating this with the given value of d,

$$\begin{aligned}
7.0313k - 10.5526 &= 3.5\\
\Rightarrow 7.0313k &= 14.0526\\
\Rightarrow k &\approx 1.9986 \approx 2
\end{aligned}$$

Rubric:

- 1 point for reducing $d = A - B$ to the single integral of $g(x) - f(x)$ over $[0, 3.75]$, without needing the crossing point.
- 1 point for the correct composite trapezoidal rule formula with $h = 0.375$.
- 2 points for the correct application of the formula to expand the integral, with the sine terms evaluated in radians and the k terms retained.
- 1 point for the correct (or an approximately correct) answer.

2. The zero y-intercept version of the linear regression model for predicting the response for a given set of n data points $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$ is $3 + 2$ (CO4) (PO1)

$$y = a_1 x$$

where a_1 is the only parameter of the regression model.

i. Derive the formula for finding the optimal value of a_1.

ii. Prove that the value of a_1 obtained using the formula derived in Question-2(i) is unique and corresponds to the absolute minimum of the optimization criterion used.

Solution:

i. The residual error for each given data point is,

$$E_i = y_i - a_1 x_i$$

So the sum of the square of the residual errors is,

$$S_r = \sum_{i=1}^{n} E_i^2 = \sum_{i=1}^{n} (y_i - a_1 x_i)^2$$

To find a_1, we look for the value of a_1 for which S_r is the absolute minimum. At first, let's do the first derivative test.

$$\frac{dS_r}{da_1} = 2\sum_{i=1}^{n}(y_i - a_1 x_i)(-x_i) = 0$$

$$\Rightarrow -2\sum_{i=1}^{n} y_i x_i + 2\sum_{i=1}^{n} a_1 x_i^2 = 0 \Rightarrow -2\sum_{i=1}^{n} y_i x_i + 2a_1 \sum_{i=1}^{n} x_i^2 = 0$$

$$\Rightarrow a_1 = \frac{\sum_{i=1}^{n} y_i x_i}{\sum_{i=1}^{n} x_i^2}$$

Rubric:

- 0.5 point for the correct equation of E_i.
- 1 point for the correct equation of S_r.
- 1.5 points for the correct derivation of a_1.

ii. To prove the uniqueness of a_1, we need to prove that its value corresponds to the absolute minimum of S_r. For this, let's invoke the second derivative test.

$$\begin{aligned}
\frac{d^2 S_r}{da_1^2} &= \frac{d}{da_1}\left(\frac{dS_r}{da_1}\right) \\
&= \frac{d}{da_1}\left(2\sum_{i=1}^{n}(y_i - a_1 x_i)(-x_i)\right) \\
&= \frac{d}{da_1}\left(\sum_{i=1}^{n}(-2x_i y_i + 2a_1 x_i^2)\right) \\
&= \sum_{i=1}^{n} 2x_i^2 > 0
\end{aligned}$$

by pragmatically assuming that all the x_i values are not zero at the same time. Hence, S_r yields the absolute minimum value at that value of a_1. So the value of a_1 that is obtained using the formula in Question-2(i) is unique. □ Q.E.D.

Rubric:

- 1.5 points for the correct second derivative test.
- 0.5 point for the correct conclusion.

3. Suppose you are analyzing the catalytic efficiency of an enzyme to determine the maximum reaction velocity $v_{\max}$ and the Michaelis constant K_m. The kinetics of the reaction can be represented by the Saturation Growth model (Michaelis-Menten equation): 4 (CO2) (PO1)

$$v = \frac{v_{\max} S}{K_m + S}$$

where v is the initial reaction velocity and S is the substrate concentration. The values of reaction velocity v and the respective substrate concentration S that you have measured experimentally are shown in the table below —

Substrate Conc., S (mM)	**Reaction Velocity,** v $(\mu mol \cdot min^{-1})$
1	1.7
2	2.9
5	5.0
10	6.7
20	8.0
40	8.9

Table 1: Experimental kinetic data for the enzymatic reaction.

Determine the value of the maximum velocity $v_{\max}$ and the Michaelis constant K_m using the data provided in Table-1. You are allowed to apply data transformation (linearization).

Solution:

Let's linearize the data at first. Taking the reciprocal of both sides of the given Saturation Growth model (Michaelis-Menten equation), we get the Lineweaver-Burk equation:

$$\frac{1}{v} = \frac{K_m}{v_{\max}} \frac{1}{S} + \frac{1}{v_{\max}}$$

Let, $z = \frac{1}{v}$, $w = \frac{1}{S}$, $a_1 = \frac{K_m}{v_{\max}}$, and $a_0 = \frac{1}{v_{\max}}$; implying that $v_{\max} = \frac{1}{a_0}$ and $K_m = a_1 \cdot v_{\max}$. The straight line that we end up with is

$$z = a_0 + a_1 w$$

Using the calculator, we obtain the following values for the $\hat{n} = 6$ transformed data points ($1/S$ and $1/v$).

$$\sum_{i=1}^{\hat{n}} w_i = 1.875, \sum_{i=1}^{\hat{n}} z_i = 1.5197, \sum_{i=1}^{\hat{n}} w_i z_i = 0.8246, \sum_{i=1}^{\hat{n}} w_i^2 = 1.3031$$

Now, let's apply the formulae for determining the corresponding linear regression parameters a_0 and a_1.

$$a_1 = \frac{\hat{n}\sum_{i=1}^{\hat{n}} w_i z_i - \sum_{i=1}^{\hat{n}} w_i \sum_{i=1}^{\hat{n}} z_i}{\hat{n}\sum_{i=1}^{\hat{n}} w_i^2 - \left(\sum_{i=1}^{\hat{n}} w_i\right)^2} = 0.4876,\ a_0 = \frac{\sum_{i=1}^{\hat{n}} z_i}{\hat{n}} - a_1 \frac{\sum_{i=1}^{\hat{n}} w_i}{\hat{n}} = 0.1009$$

Hence, we obtain

$$\begin{aligned} v_{\max} &= \frac{1}{0.1009} \\ &= 9.9108 \end{aligned}$$

$$K_m = 0.4876 \times 9.9108 = 4.8325$$

Plugging these values into the given Saturation Growth model equation, we get,

$$v = \frac{9.9108S}{4.8325 + S}$$

Rubric:

- 1 point for the correct linear representation.
- 1 point for the correct a_1 formula.
- 0.5 point for the correct a_1 value.
- 1 point for the correct a_0 formula.
- 0.5 point for the correct a_0 value.
- 0.5 point for the correct $v_{\max}$ value.
- 0.5 point for the correct K_m value.

4. What is Regression Analysis? Provide a precise formulation of the problem statement of a regression model. 1 (CO3) (PO1)

Solution:

In statistical modeling, regression analysis is a set of statistical processes for estimating the relationships between a dependent variable and one or more independent variables.
Formally, the problem statement for a regression model is:

> Given n data points $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, the goal is to determine the function $y = f(x)$ that is the best-fit predictor function of the underlying data distribution.

Rubric:

- 0.5 point for a correct definition.
- 0.5 point for a correct problem formulation.